☐ N-N scattering length $a \approx 5-25$ fm

Cross section $\sigma = 4\pi a^2$ fm^2

$= 4\pi \times 5^2 - 4\pi \times 25^2$ fm^2

$= 314 - 7850$ fm^2

(∵ 1 fm^2 = 10^{-30} m^2 and 10^{-28} m^2 = 1 barn = 100 fm^2)

So $\boxed{\sigma = 3.14 - 78.5 \text{ barn}}$

Radioactive Decay

There are few radioactive elements (examples are given below) from where spontaneous emission of α and/or β and/or γ rays are observed.

- α particle is basically $_2He^4$ nucleus
- β " " " electron
- γ " " " photon (with high frequency, higher than x-ray)

Examples:

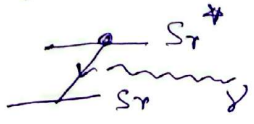
- Radioactive uranium element:

$$_{92}U^{238} \longrightarrow {}_{90}Th^{234} + (_2He^4) \rightarrow \alpha \text{ particle}$$

- Carbon dating via β ray observation

$$_6C^{14} \longrightarrow {}_7N^{14} + (e^-) \rightarrow \beta \text{ particle}$$

- $^*_{38}Sr^{87} \longrightarrow {}_{38}Sr^{87} + (\gamma) \rightarrow \gamma$ ray



Quantifying radioactivity: Activity of substance is measured by the no. of atoms that disintegrate per second.

It follows (radioactive) exponential decay law: $\boxed{N(t) = N_0 e^{-\lambda t}}$, where λ = decay constant

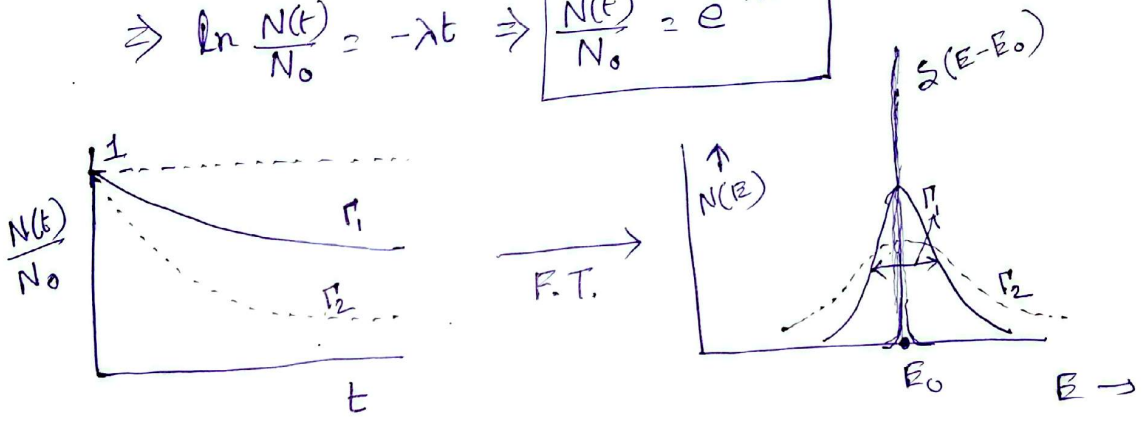
N_0 = no. at $t = 0$.

Based on probability hypothesis

$$\frac{dN(t)}{dt} \propto -N(t) \Rightarrow \frac{dN}{dt} = -\lambda N \Rightarrow \int_{N=N_0}^{N=N(t)} \frac{dN}{N} = -\int_{t=0}^{t=t} \lambda\, dt$$

i.e $\ln N(t) - \ln N_0 = -\lambda t$

$\Rightarrow \ln \frac{N(t)}{N_0} = -\lambda t \Rightarrow \boxed{\frac{N(t)}{N_0} = e^{-\lambda t}}$



$\psi(t) \sim e^{-iE_0 t}$ for stationary matter wave function

$\psi(t) \sim e^{-iE_0 t} e^{-\Gamma t/2}$ for decaying matter wave function

$$\psi(E) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \psi(t) e^{iEt}\, dt$$

$$= \frac{1}{\sqrt{2\pi}} \frac{i}{(E-E_0) + i\Gamma/2}$$

↓ Probability in time axis

$N(t) \sim |\psi(t)|^2 \sim e^{-\Gamma t}$

↓ probability in energy axis

$$N(E) \sim \frac{1}{(E-E_0)^2 + \Gamma^2/4}$$

Discovery of neutrino particle (ν)

1896-1900 : Becquerel (Nuclear Physicist) discovered radioactive uranium salts. Using magnetic field deflection and measuring charge to mass ratio, Becquerel confirmed that β-rays is rays/flow of electrons.

1914 - 1927 : Chadwick and other nuclear physicists experimentally observed - a continous β-ray spectrum.

If we explain β-decay process as

$$n \rightarrow p + e^-$$

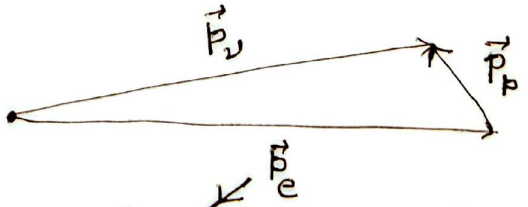
$-\vec{p}_{cm}$ $\quad$ $+\vec{p}_{cm}$

then instead of continous β-ray spectra, an end point energy is expected, which was not observed in experiment.

1930 : Resolved this problem by theoretician Pauli.

β-decay process is re-explained as

$$n \rightarrow p + e^- + \bar{\nu}_e$$



A new particle $\bar{\nu}_e$ is proposed which will compensate and explain the continous spectra of β-ray.

It can be larger than end point energy and continous spectra

Accelerator:-

Linear accelerator:

One electrode

$$\frac{1}{2}mv_1^2 = eV$$

$$\Rightarrow \boxed{v_1 = \sqrt{\frac{2eV}{m}}}$$

n linear electrode

$$\frac{1}{2}mv_n^2 = neV$$

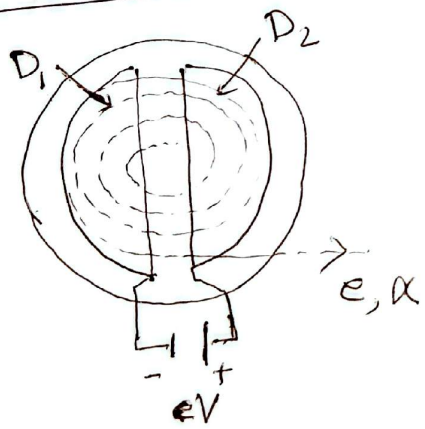
$$\Rightarrow \boxed{v_n = \sqrt{\frac{2neV}{m}}}$$

$$\boxed{v_n = \sqrt{n}\, v_1}$$

By increasing 'n', velocity of electrons or any projectile/accelerating particles can be increased.

Limitation: Long space with vacuum facility is required.

Cyclotron:



$$\frac{mv^2}{r} = evB$$

$$\Rightarrow \boxed{r = \frac{mv}{eB}}$$

$\Rightarrow$ Angular velocity

$$\boxed{\omega = \frac{v}{r} = \frac{eB}{m}}$$ → Also called cyclotron angular frequency

cyclotron time period

$$\boxed{T_B = \frac{2\pi}{\omega} = \frac{2\pi m}{eB}}$$

Maximum energy $E_{max} = \frac{1}{2}mv_{max}^2 = neV$, n = no of crossing Disc

$$= \frac{1}{2}\frac{B^2e^2r_{max}^2}{m}$$